Step-1 kill statistics and asymptotic locality:

empirical evidence for survivor fraction $\to 0$

(satellite for the locality program)

Abstract

At n external legs, the layer-0 ("Step-1") hidden-zero kill mechanism forces $a_M = 0$ for the coefficient of every chord multiset M of size $n-3$ that satisfies the killability conditions (K1)+(K2) at some cyclic zone. Through $n = 9$ we report the exact count of surviving (non-killable) non-triangulation multisets at each n, the resulting kill rate, and the cyclic-orbit structure of the survivors. The data show: (i) the kill rate on non-triangulations exceeds 99.6% for every $n \geq 7$ measured; (ii) the kill rate is monotonically increasing in n; (iii) at $n = 9$ already 99.888% of all non-triangulation multisets are killed by Step-1 alone. We state the precise asymptotic conjecture (Item 4 of the master plan) and identify what a proof would need to establish. This satellite is companion to the local-survival lemma of `paper/step-one-doesnt-kill-triangulations/`, which closes the matching combinatorial side (Item 2).

1 Setup

We adopt the conventions of the root README: n vertices labelled $1, \ldots, n$ cyclically; planar Mandelstam invariants $X_{ij} = (p_i + \cdots + p_{j-1})^2$ for chord pairs (i, j); rational ansatz

$$B \;=\; \sum_M \frac{a_M}{\prod_{c \in M} X_c}$$

where M runs over size-$(n-3)$ chord multisets. The cyclic hidden-zero zone $\mathcal{Z}_{r,r+2}$ has special chord $X_{r,r+2}$, bare chord $X_{r+1,r-1} = -X_{r,r+2}$, and substitutions $X_{r+1,k} = X_{r,k} - X_{r,r+2}$ for k in cyclic range.

$M \in \mathcal{K}_{r,r+2}$ ("killable at zone r") iff

(K1) M contains neither the special nor the bare;

(K2) for every (companion, substitute) pair $(X_{r,k}, X_{r+1,k})$, M does not contain both members.

A multiset M is a *Step-1 survivor* iff it is in no $\mathcal{K}_{r,r+2}$ for any r. By the local-survival lemma of [1], every triangulation T is a Step-1 survivor; the non-triangulation Step-1 survivors are the focus of this note.

2 The data

The script `computations/step1_layer0_kill/kill_enumeration/step1_uncaught.py` enumerates all size-$(n-3)$ multisets, applies the (K1)+(K2) test at every cyclic zone, and reports the survivors. The output through $n = 9$:

n	total multisets	triangulations C_{n-2}	non-tri total	non-tri survivors	non-tri kill %
4	2	2	0	0	—
5	15	5	10	0	100 %
6	165	14	151	0	100 %
7	2 380	42	2 338	7	99.700 %
8	42 504	132	42 372	100	99.764 %
9	906 192	429	905 763	1 011	99.888 %

Triangulation counts match the Catalan numbers C_{n-2} exactly, consistent with the local-survival lemma of [1]. The total multiset count at n is $\binom{d+n-4}{n-3}$ where $d = n(n-3)/2$.

Kill rate is monotonically increasing. For every $n \geq 7$ in the data, the kill rate strictly exceeds the rate at $n-1$: 99.700% $\to$ 99.764% $\to$ 99.888%. The headline number at $n = 9$ is that fewer than 0.12% of all non-triangulation multisets escape Step-1.

3 Cyclic orbit structure of the survivors

The survivors at each n are closed under the cyclic-rotation action $v \mapsto v + 1 \pmod{n}$ on chord endpoints, so they break into $\mathbb{Z}_n$-orbits of size $\leq n$.

***n* = 7.** All 7 survivors form a single $\mathbb{Z}_7$-orbit (the "fish" family); they are the cyclic shifts of $M_1 = \{(1,3),(1,6),(2,4),(4,6)\}$. See `paper/Laurent series for hard kills/cascade_n7.tex` for the kill.

***n* = 8.** The 100 survivors break into multiple $\mathbb{Z}_8$-orbits (orbit-shape detail in `computations/step1_layer0_kill/` The companion script `computations/step3_laurent/cascade_n8/cascade_kill_n8.py` verifies **every one** of these 100 survivors dies at Step 3 via a depth-1 Laurent cascade — 90 of them at order X_*^{-3} (leading + 1) and 10 at order X_*^{-4} (leading + 2), with all cousins step-1 killable at one neighbouring zone. So at $n = 8$, the cumulative kill rate of **Step-1 + Step-3** on non-triangulations is 100%.

***n* = 9.** 1011 survivors. Cyclic-orbit decomposition is in the script's PDF output. The Step-3 cascade verification at $n = 9$ has not yet been run.

4 Asymptotic conjecture (Item 4)

Let $S(n)$ denote the number of non-triangulation Step-1 survivors at n, and $T(n)$ the total number of non-triangulation size-$(n-3)$ multisets ($T(n) = \binom{d+n-4}{n-3} - C_{n-2}$). The data $(S(n), T(n))$ for $n = 5, \ldots, 9$:

n	$S(n)$	$T(n)$	$S(n)/T(n)$	$S(n+1)/S(n)$
5	0	10	0 %	—
6	0	151	0 %	—
7	7	2 338	0.300 %	—
8	100	42 372	0.236 %	≈ 14.3
9	1 011	905 763	0.112 %	≈ 10.1

The ratio $S(n)/T(n)$ is roughly halving per increment of n, while $S(n+1)/S(n)$ is decreasing (suggesting sub-exponential survivor growth). The total grows superexponentially.

Conjecture 1 (Asymptotic locality from Step-1 alone)**.** *There exist constants $C, \alpha > 0$ such that for every $n \geq 7$,*

$$\frac{S(n)}{T(n)} \leq \frac{C}{n^\alpha} \quad \textit{(polynomial decay)} \qquad \textit{or} \qquad \frac{S(n)}{T(n)} \leq C \cdot 2^{-n} \quad \textit{(exponential decay).}$$

The data favour the exponential form with rate $\sim 2^{-n}$ but the polynomial form is not ruled out by $n \leq 9$.

A proof would need to:

1. identify the structural family of multisets that satisfies neither (K1) nor (K2) at every zone (the survivors);
2. bound the cardinality of that family by a closed-form expression in n;
3. compare to $T(n)$, whose growth is known.

The cyclic-orbit structure reduces (1)–(2) to characterising one representative per orbit, then multiplying by orbit size $\leq n$. At $n = 7$ all 7 survivors are one orbit; at $n = 8$ the orbit count is the natural quantity to track in the proof. We do not pursue this here.

What this conjecture is *not* needed for. The all-n locality theorem does *not* depend on Conjecture 1. What it depends on is the per-survivor Laurent cascade (Item 10 of the master plan): every Step-1 survivor dies via a depth-1 cascade with prerequisites all of layer-0 type. Verified at $n = 7$ symbolically and at $n = 8$ in `computations/step3_laurent/cascade_n8/`. The asymptotic conjecture here is supportive evidence, not a load-bearing component.

5 What the satellite Step-1 ecosystem now looks like

The two satellites that close out the Step-1 analysis:

- `paper/step-one-doesnt-kill-triangulations/` — proves the *local* side: no triangulation is ever Step-1 killable. Closes Item 2 of the master plan.
- `paper/step-one-kill-statistics/` (this note) — documents the *global* kill rate: which fraction of non-triangulations does Step-1 catch, and how does that fraction grow with n? Backs Item 4 of the master plan.

Together with the homogeneity-decomposition lemma (`paper/B=0->B_i = 0/`, Item 1) and the Laurent-cascade analysis (`paper/Laurent series for hard kills/`, Item 10), Step-1 is fully understood up to the asymptotic question above.

References

[1] J. Valenzuela, *Layer-0 hidden-zero kills no triangulation*, `paper/step-one-doesnt-kill-triangulations/triangulation_layer0.pdf`.

[2] J. Valenzuela, *The Laurent-tail kill at* $n = 7$, `paper/Laurent series for hard kills/cascade_n7.pdf`.

[3] A. Rodina, *Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in* $\mathrm{Tr}(\phi^3)$ *theory*, arXiv:2406.04234.